

KitPri v4 — Cooking Sound Detection

Project report: dataset rebuild, model selection, distillation, threshold tuning, and quantization for on-device deployment on the Samsung Telegram bot.

1. What This Project Is

The goal was to build a small AI model that listens to short audio clips and decides whether "cooking sounds" are happening or not — running on a Samsung device inside a Telegram bot. The earlier version (v2) of this model was missing quiet, easy-to-miss sounds like electrical hums, so this round (v4) rebuilt the dataset and the model from the ground up to fix that.

2. The New Dataset (v4)

Instead of clean, single sounds, the new dataset mixes 2 to 4 different sounds together in every clip, at different volumes, so it looks more like real-world audio. This makes it far more realistic, but also much harder for a model to learn from.

Total clips	7,200
Training clips	6,300
Validation clips	450
Test clips	450
Clip make-up	Synthetic mix of 2–4 layered sounds per clip

3. Choosing the Right Model: EfficientNet-B0 vs. AST

EfficientNet-B0 (the model that had worked well on the older, easier dataset) was tried six different ways — different freezing strategies, learning rates, dropout settings, augmentation, and mixup. Every attempt landed in the same narrow band. That consistency was the signal: the problem wasn't tuning, it was that this architecture isn't well suited to this harder, overlapping-sound data.

AST (Audio Spectrogram Transformer) — a model built specifically for audio using attention rather than image-style filters — was tried next, and it won clearly, in a single epoch of training. AST became the teacher model for the rest of the project.

Model	Val F1	Test F1	Test Precision	Test Recall	Test Accuracy	Params
EfficientNet-B0 (best of 6 runs)	~0.68	~0.63–0.68	—	—	—	~5.3M
AST (teacher, epoch 1)	0.8233	0.8129	0.8462	0.7822	82.0%	86.2M

AST training detail: best result came at epoch 1 (val F1 = 0.8233); after that the model began overfitting (training F1 climbed past 0.98 while validation F1 fell), so training was stopped early at epoch 6 and the epoch-1 checkpoint was kept.

4. Why Some Clips Are Genuinely Hard

Looking into the clips both models struggled with, we found two contributing issues:

- **Overlapping audio ("clip_type C"):** some clips labeled "cooking" also contain a non-cooking sound playing at the same time — sometimes louder than the cooking sound. The model has to pick out a quiet cooking sound while something else competes for attention in the same clip. Accuracy on these overlapping clips dropped sharply (to roughly 39–50%), while clean, non-overlapping clips scored much better (75–78%). This difficulty is by design — it reflects real-world messiness the model needs to learn to handle.
- **Messy negative examples:** the "not cooking" training clips came from one large, unstructured pool of 1,160 unrelated sounds (from the ESC-50 dataset) — dogs barking, glass breaking, and more, all lumped together with no organization. This makes "not cooking" a harder, messier category to learn than "cooking," which was cleanly organized into 10 clear subtypes.

5. Shrinking It Down: Distillation into MobileNetV2

AST is too large to run on the Samsung device (86.2 million parameters). To get a model small enough to deploy, AST was used as a "teacher" to train a much smaller "student" model — MobileNetV2 (2,225,153 parameters, roughly 40x smaller) — by having the student learn to imitate the teacher's predictions, in addition to learning from the true labels directly (temperature = 3.0, alpha = 0.4).

Training ran for 14 epochs before early stopping. The best checkpoint was epoch 8, with the student's training F1 continuing to climb well past validation F1 in later epochs — a sign of overfitting, which is why the epoch-8 checkpoint (not a later one) was kept as final.

Metric	Validation	Test
F1 score	0.7662	0.7226
Precision	0.7468	0.7598
Recall	0.7867	0.6889
Accuracy	76.00%	73.56%

Student model size (FP32, before compression): 8.49 MB. This is the "student_mobilenet_fp32" checkpoint — the original, un-quantized model, and the largest/most accurate version produced.

6. Deciding the Cutoff: Threshold Tuning

By default, the model calls a clip "cooking" if it's more than 50% confident. Sweeping other cutoff points on the validation set (never the test set, to avoid leaking information into the final check) found that a slightly lower confidence bar gives a better balance of catching real cooking sounds without too many false alarms.

Chosen decision threshold: **0.44** confidence (compared with 0.40 that had been used for the earlier v2 model).

7. Shrinking the File: INT8 Quantization

To fit comfortably on the Samsung device, the model's numbers were compressed from precise decimals (FP32) into smaller integers (INT8) — cutting file size and speeding up inference. An earlier attempt at this (a different, "dynamic" quantization method) had broken the model outright, collapsing its accuracy to near-zero, because it only quantizes weights and leaves a mismatch with real activation ranges at inference time.

This time, **static quantization with calibration** was used instead — the correct method for CNN-style models like MobileNetV2. A calibration pass over 500 real training clips let the model observe realistic activation ranges before conversion, which is the step that was missing before. The result was clean: no collapse, and a well-understood, acceptable accuracy trade-off.

Metric	FP32 (original)	INT8 (quantized)	Change
Val F1	0.7662	0.7314	−0.0349
Test F1	0.7226	0.6910	−0.0316
Test Precision	0.7598	0.7634	+0.0036
Test Recall	0.6889	0.6311	−0.0578
Test Accuracy	73.56%	71.78%	−1.78 pts
Speed (ms/clip, CPU)	57.31 ms	25.51 ms	2.25x faster
File size	8.49 MB	2.80 MB	−67.0%

Quantization used 500 calibration clips sampled from the training set. Final INT8 file confirmed well under the device's 60 MB memory budget.

8. Bottom Line

The final deployed model is a 2.80 MB INT8 MobileNetV2 that runs 2.25x faster than its FP32 version, fits comfortably inside the Samsung device's 60 MB memory budget, and keeps most of the accuracy of a much larger, more capable AST teacher model — with a clear, evidence-based explanation for where its remaining errors come from (overlapping-sound clips that are intentionally the hardest cases in the dataset).

Stage	Test F1	Size	Notes
AST (teacher)	0.8129	~345 MB (86.2M params)	Too large for device
MobileNetV2 student, FP32	0.7226	8.49 MB	Distilled from AST
MobileNetV2 student, INT8 (final)	0.6910	2.80 MB	Deployed to Samsung device